



The Steam Marketplace, 2004-2018

Pricing, popularity, and quality across 26,643 real Steam releases — and what they reveal about how the PC games market industrialized.

One dataset. Twelve questions. One story.

01

Choose a real dataset

26,643 Steam games (2004–2018), scraped from Steam/SteamSpy via TidyTuesday — real-world, rich, and varied.

02

Ask analytical questions

12 multi-dimensional questions relating price, ownership, playtime, quality, and time.

03

Build publication visuals

12 Plotly figures — CVD-safe, decluttered, explanatory titles — in a Jupyter notebook.

04

Ship an interactive dashboard

A 4-tab Streamlit app with live filters, deployed on Streamlit Community Cloud.

THE DATASET

26,643 games. Four attribute types. Fifteen years.

Source: Steam Store + SteamSpy scrape, distributed via the R for Data Science community's TidyTuesday project (week of 2019-07-30). Real-world data, not a synthetic teaching set.

Attribute type	Columns
Numerical	price, average_playtime, median_playtime, metaspcore, owners_est
Categorical	developer, publisher, owners (ordinal bracket), price_tier
Temporal	release_date, release_year, release_month
Text	game (title) — length, word count, sequel/edition markers

26,643

games analyzed

2004-2018

release-year span

~10.7%

have a Metacritic score

From raw scrape to analysis-ready table

1

Parse & filter

Parsed `release_date`; kept 2004–2018 full calendar years to avoid partial-year 2019 bias.

2

Fix a hidden bug

The `owners` column used non-breaking spaces (`\xa0`) instead of regular spaces — silently broke every category match until caught and normalized.

3

Engineer features

`price_tier` (5 bins), `owners_est` (bracket midpoint), title text stats, developer/publisher catalog size, playtime-skew index.

4

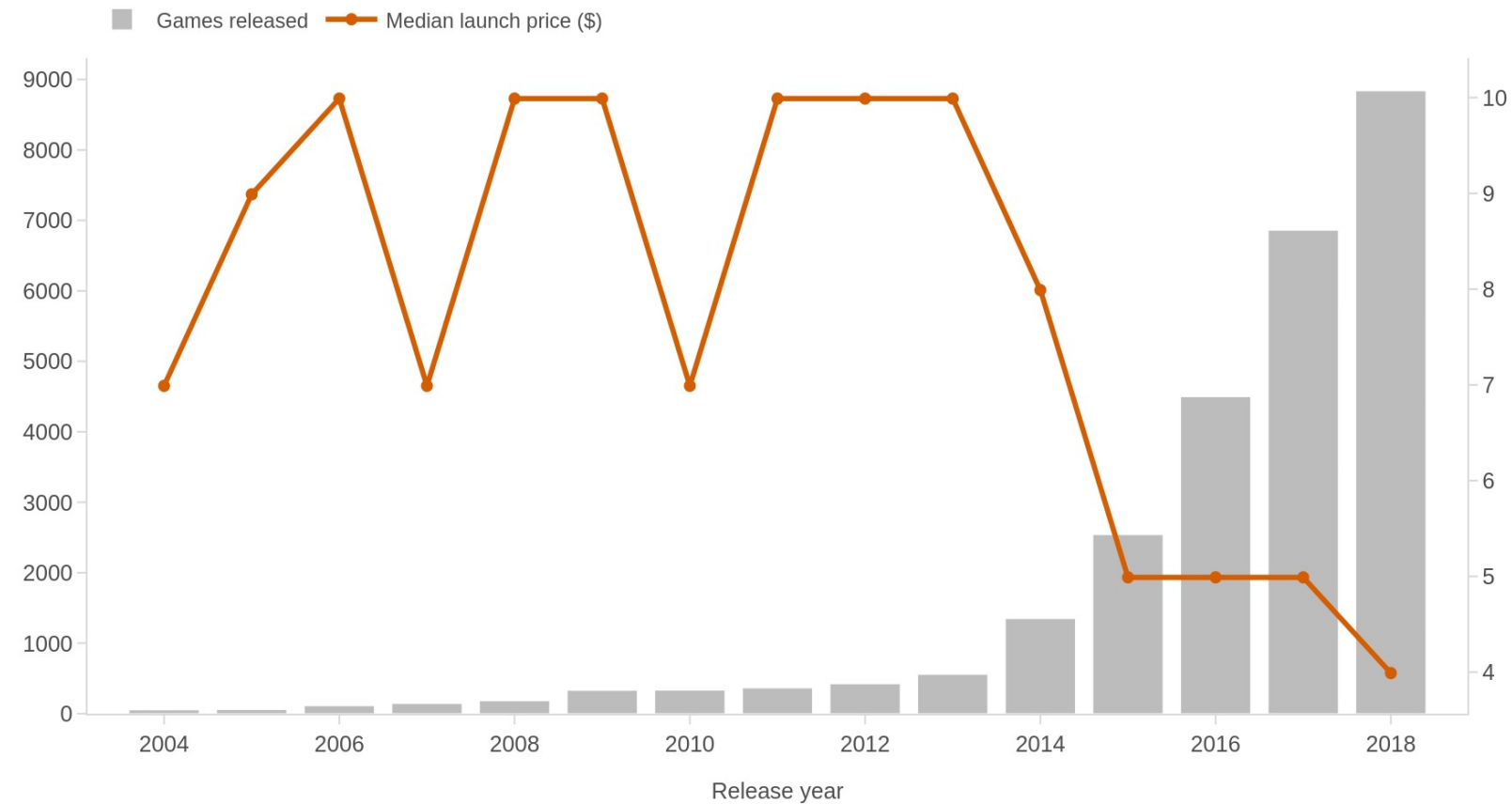
Handle missingness honestly

`metascore` = 0 recoded as missing (not “bad”); `has_critic_score` flag added instead of silently imputing.

Releases grew 30x while the typical price was cut in half

Releases grew 30x while the typical price was cut in half

Annual Steam releases (bars) vs. median launch price (line), 2004–2018



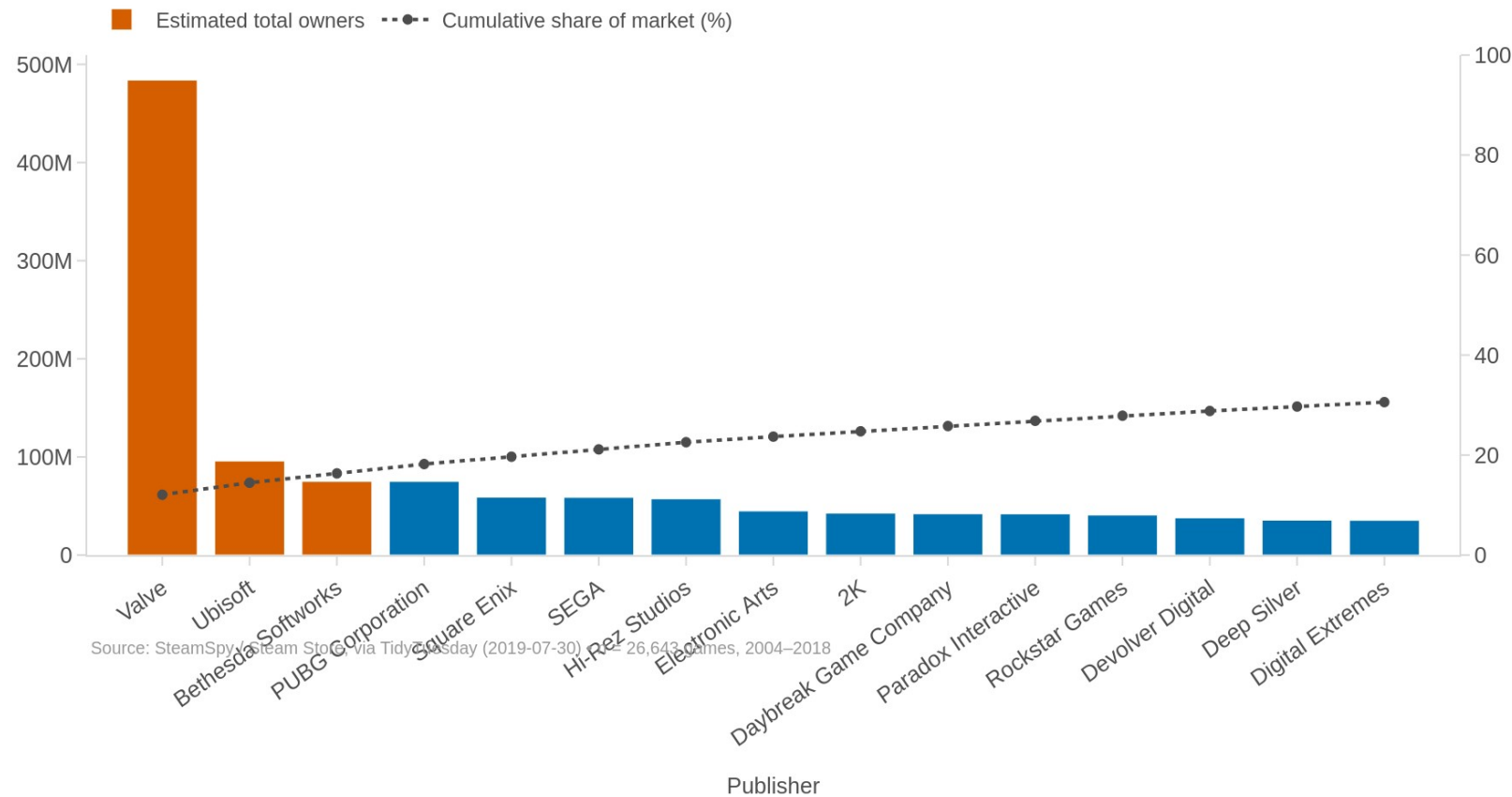
KEY TAKEAWAY

Annual Steam releases exploded from dozens in the mid-2000s to nearly 9,000 by 2018, driven by Greenlight/Direct opening the platform to indie developers. Median launch price fell from ~\$10– 15 to under \$6 over the same period — the market shifted from scarce & premium to abundant & cheap.

15 publishers (of ~14,000) control a huge share of the market

Just 15 publishers (of ~14,000) account for a huge share of all owners

Top 15 publishers by estimated total owners, with cumulative market share



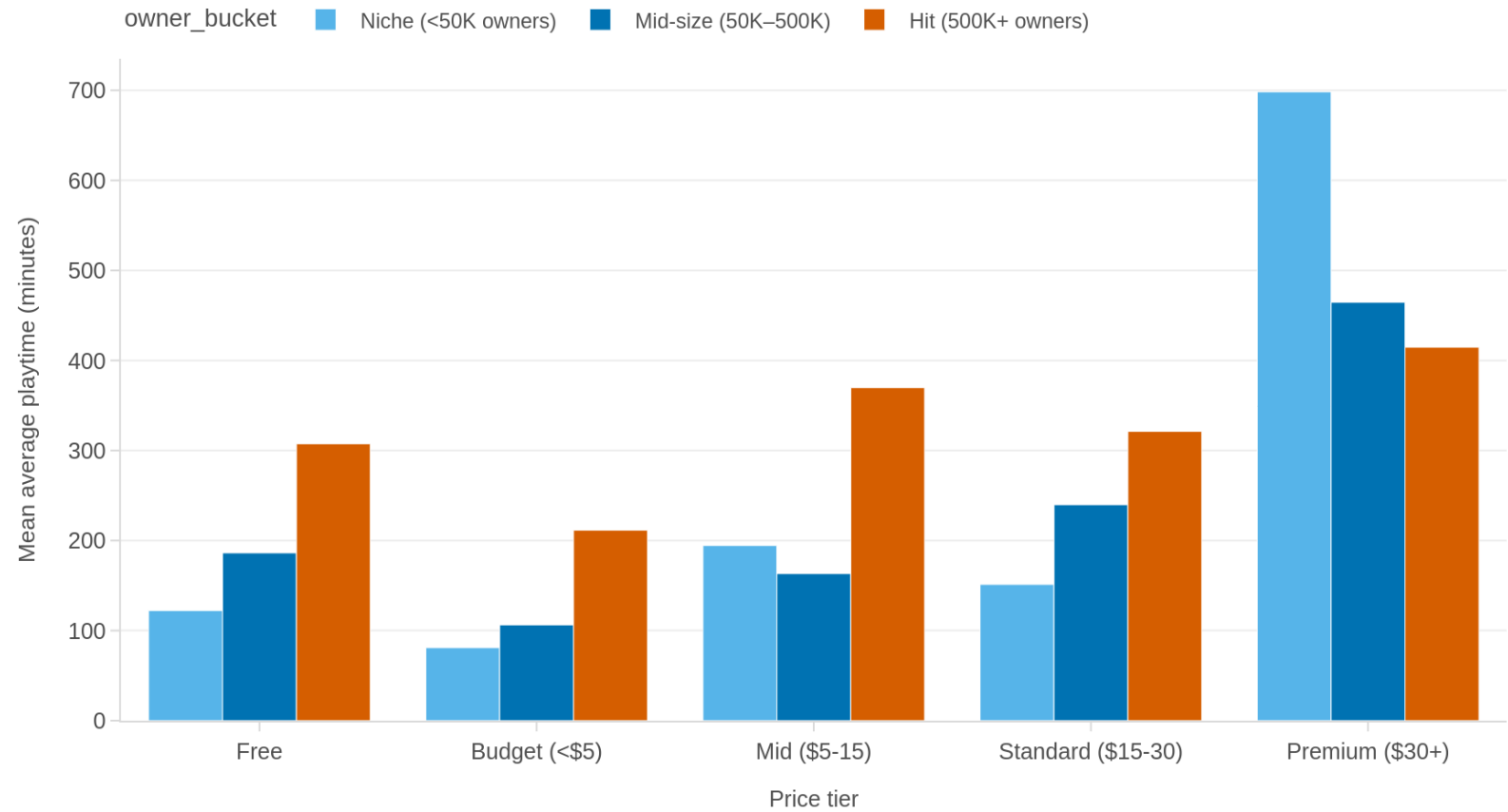
KEY TAKEAWAY

This is a classic long-tail market. Valve and a handful of major publishers hold a disproportionate share of total estimated ownership, while tens of thousands of small publishers each hold a sliver of the pie.

Popularity — not price — drives playtime

Hit games hold players' attention far longer than price alone predicts

Mean average playtime (minutes) by price tier and popularity tier — games with recorded playtime only

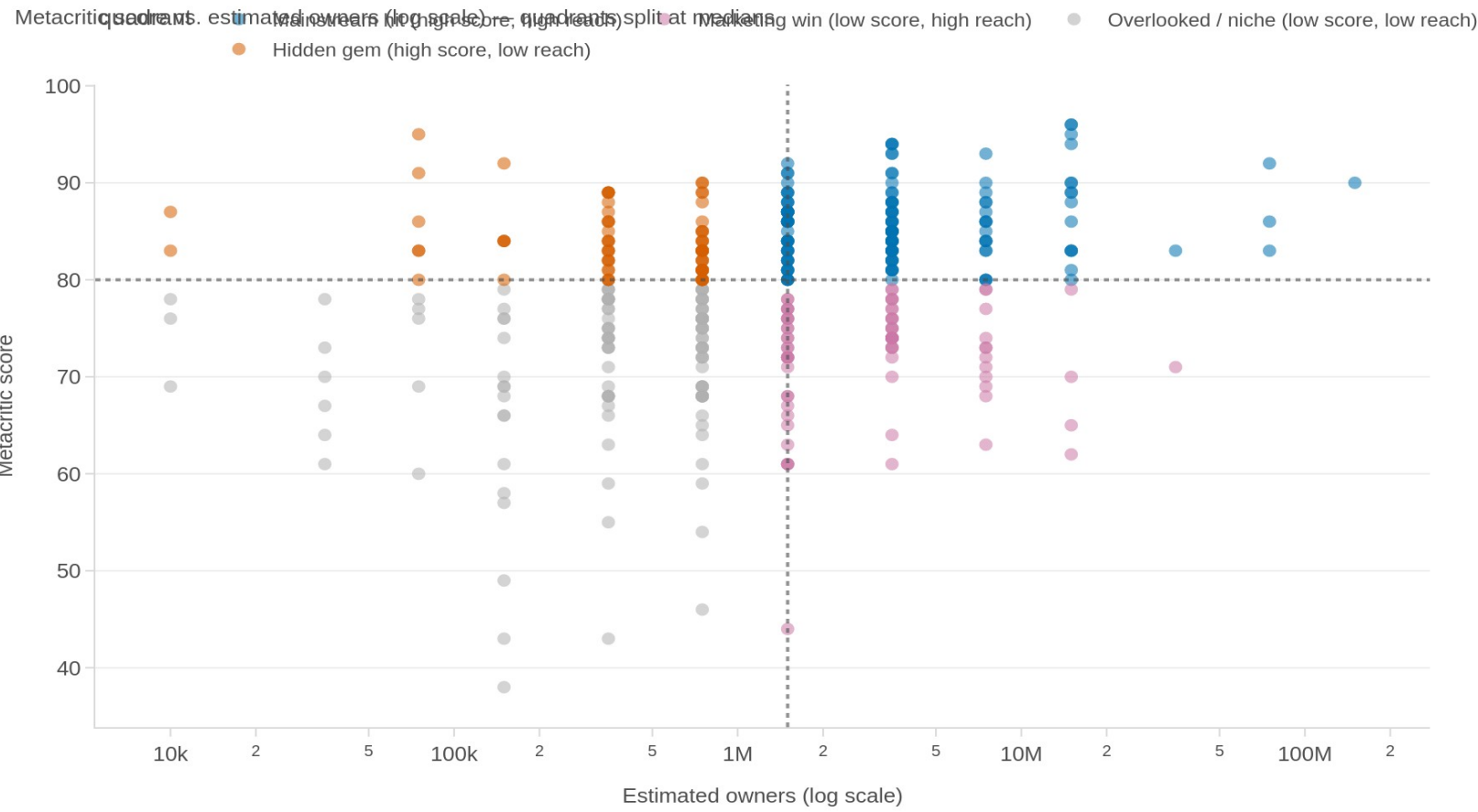


KEY TAKEAWAY

Within every price tier, “Hit” games rack up far more average playtime than niche titles at the same price point. Paying more does not reliably buy more engagement — popularity dominates.

A cluster of 'hidden gems' scores well but reaches few players

A distinct cluster of 'hidden gems' scores well but reaches few players



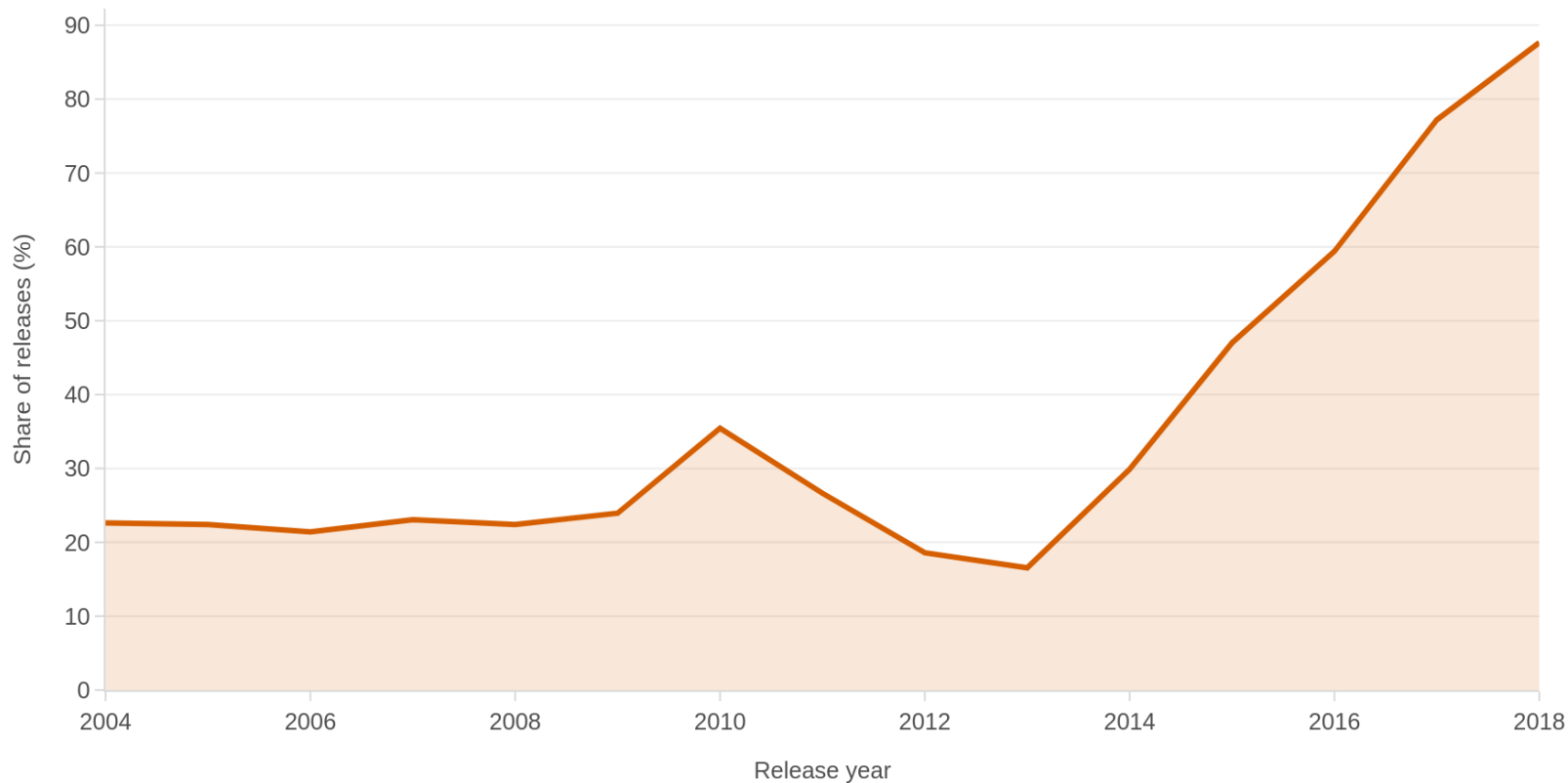
KEY TAKEAWAY

A meaningful share of critically strong games never find a large audience. Steam's discovery problem is real — quality alone is far from sufficient for commercial reach.

By 2018, over 8 in 10 new releases failed to reach 20K owners

By 2018, over 8 in 10 new releases failed to reach 20,000 owners

Share of each year's releases in the lowest ownership bracket (<20K estimated owners)



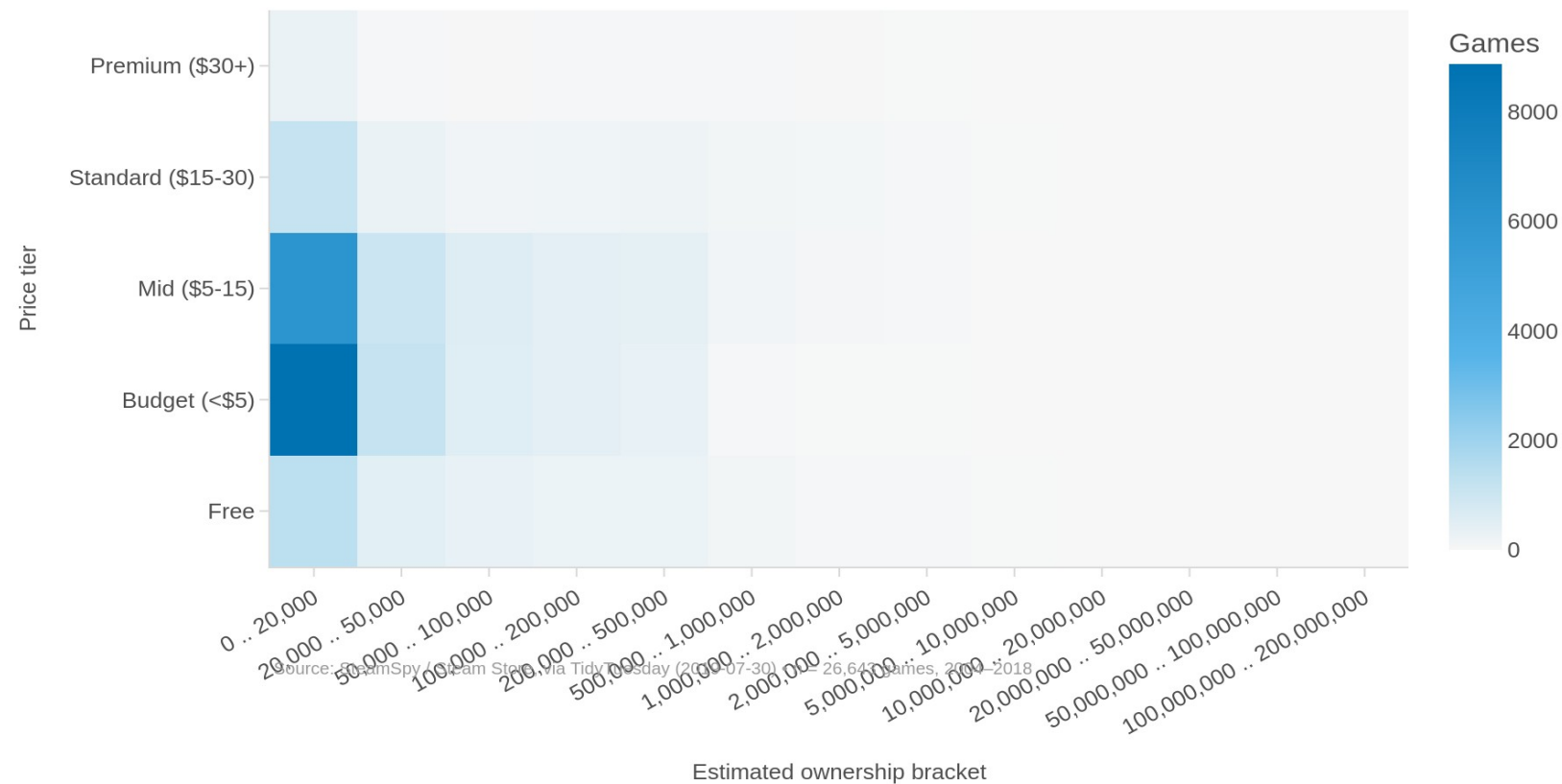
KEY TAKEAWAY

As releases multiplied post-2014, the share of games stuck in the lowest ownership bracket rose sharply. More releases chasing a similarly sized player base means the median new game is increasingly likely to go unnoticed.

Cheap-and-niche is the default state of the catalog

Cheap-and-niche is the default: most games sit in the bottom-left cell

Number of games by price tier x estimated ownership bracket (2004–2018)



KEY TAKEAWAY

The overwhelming majority of games sit in the “Budget/Free × under 20,000 owners” cell. Commercial hits appear at every price tier, but they are rare outliers against a backdrop of inexpensive, low-visibility releases.

Explore it yourself — live on Streamlit Community Cloud



Market Overview

Volume/price trend, long-tail growth, publisher concentration



Pricing & Engagement

Price vs. playtime, audience polarization, sequel premium



Quality & Discovery

Hidden gems quadrant, developer scale vs. quality, score-playtime curve



Explore the Data

Price×popularity heatmap, sortable table, CSV download

Every chart on every tab responds live to sidebar filters: release-year range, price tier, popularity tier, Metacritic-reviewed toggle, and publisher search.

Live URL: [<paste your Streamlit Community Cloud URL here after deploying>](#)

Source repo: [<paste your public GitHub repo URL here>](#)

CONCLUSIONS

What twelve questions on one catalog add up to



The market industrialized.

Releases grew ~30× while median prices fell by more than half — curated & premium became open & high-volume.



Popularity beats price.

Hit games out-engage niche titles at every price point; paying more doesn't reliably buy more playtime.



It's winner-take-most.

A handful of publishers dominate ownership, sequels command a persistent premium, and hidden gems are common.



Abundance has a cost.

As releases multiplied post-2014, the share of games that never reach 20,000 owners rose sharply.

LIMITATIONS

- Ownership & playtime are SteamSpy estimates — subject to API changes and player privacy settings.
- Metacritic coverage is sparse (~10.7% of titles), so quality-based findings lean on a smaller subset.
- Snapshot ends in 2019 — predates recent shifts in Steam's discovery algorithm and the 2020s indie boom.

Thank you

Full analysis, code, and data cleaning steps: [notebook/Final_Project_Analysis.ipynb](#)

Live dashboard: [<Streamlit Community Cloud URL>](#) • Source: [<GitHub repo URL>](#)